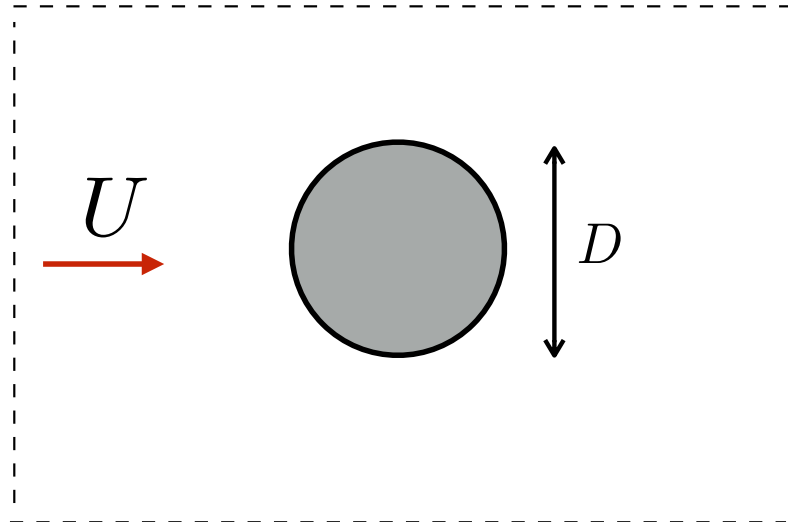


The “flow past a circular cylinder”



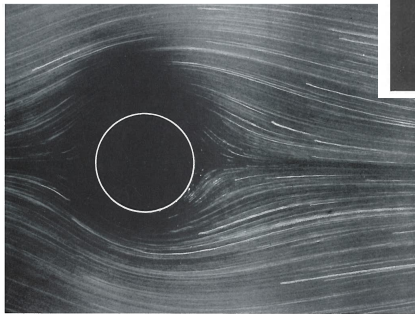
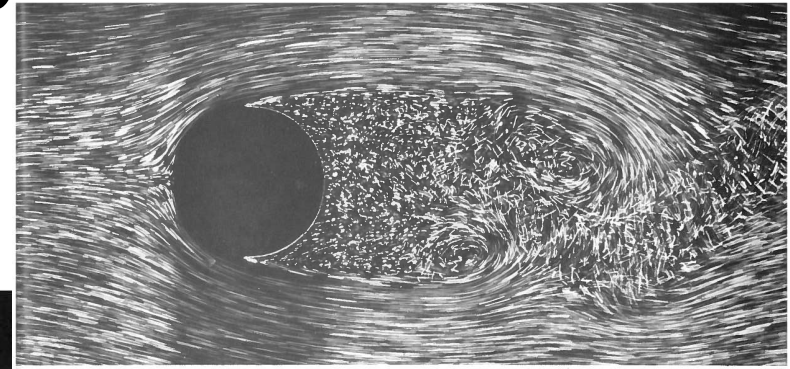
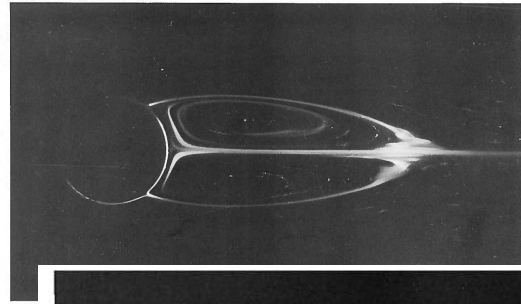
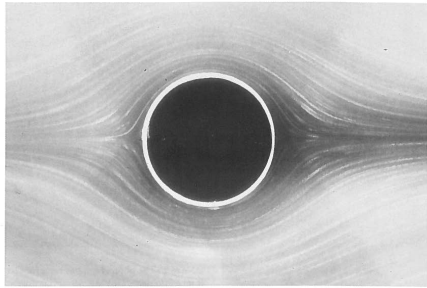
- A canonical, **two-dimensional**, **steady (or unsteady)** **incompressible** flow
- In principle, fully characterized by the Reynolds number

$$\text{Re} = UD/\nu$$

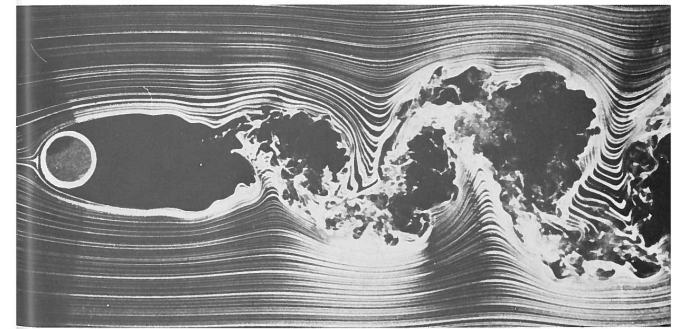
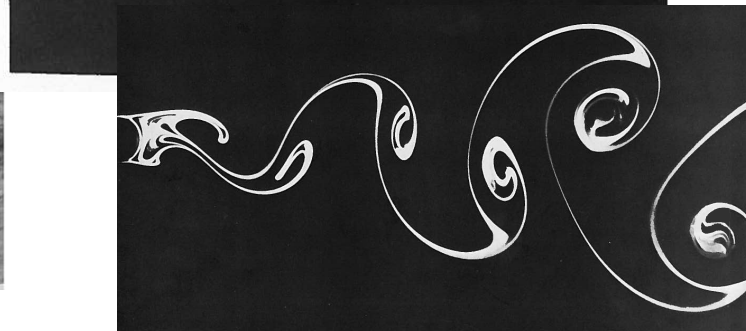
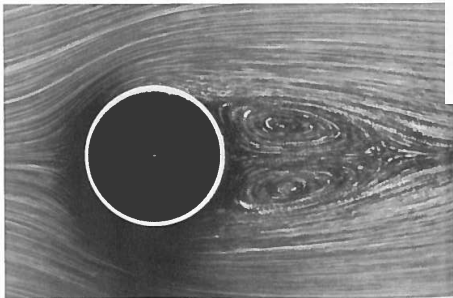
An Album of Fluid Motion

- All images that follow are taken from Milton Van Dyke's "An Album of Fluid Motion".

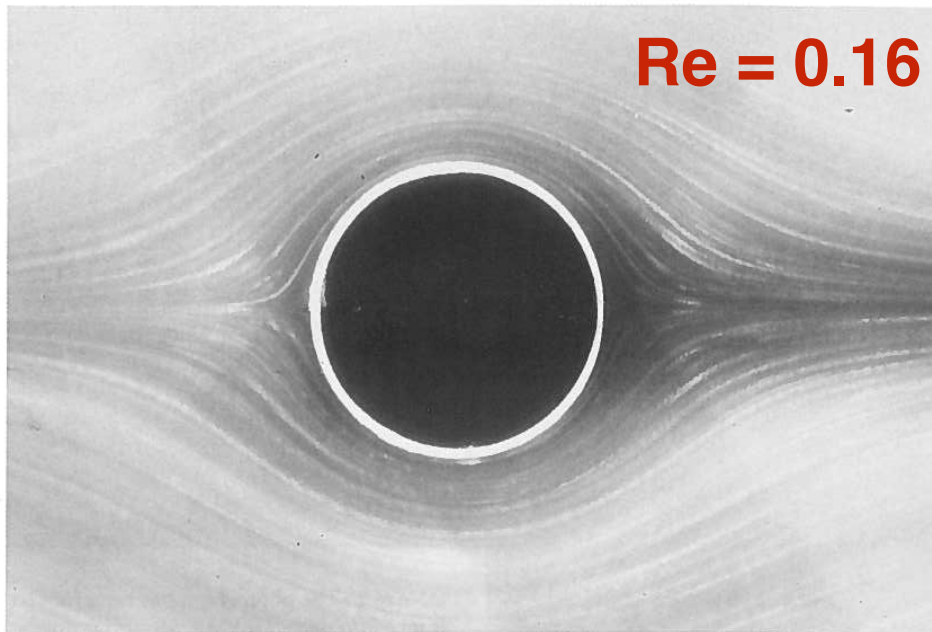
Know thy flow!



What distinguishes these flows around a cylinder?



The flow depends on Re

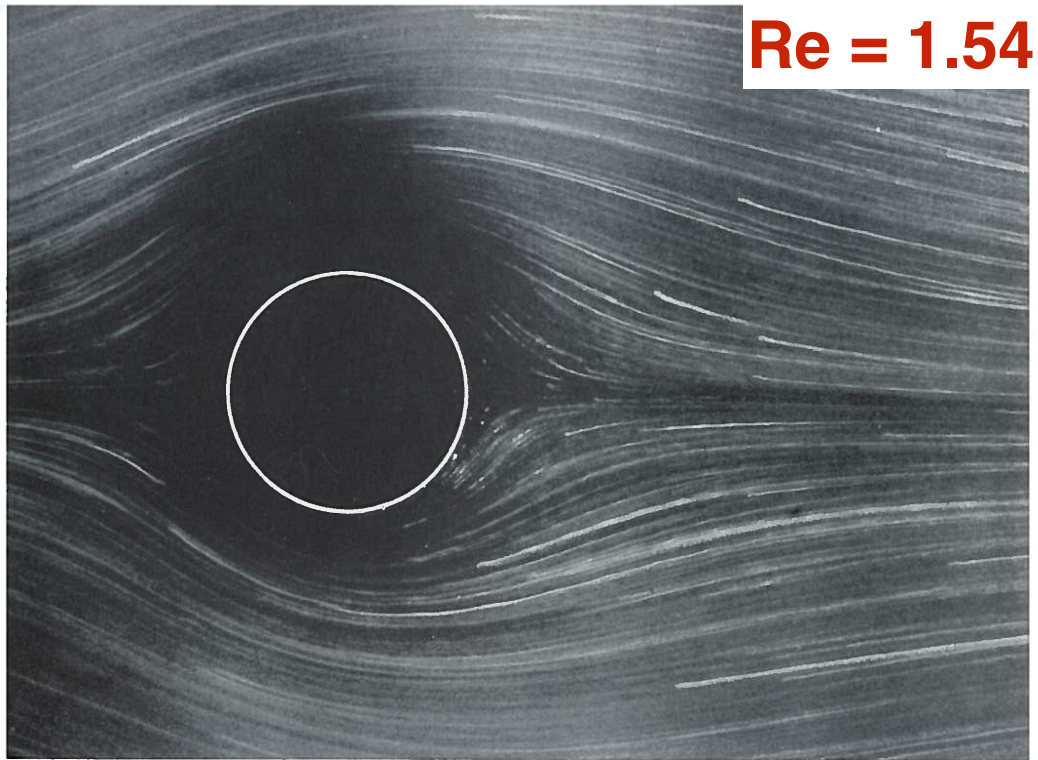


6. Uniform flow past a circular cylinder at $R=0.16$. That the flow is from left to right can scarcely be deduced from the streamline pattern, because in the limit of zero Reynolds number the flow past a solid body is reversible, and hence symmetric about a symmetric shape. It resem-

bles superficially the pattern of potential flow in figure 1, but the disturbances to the uniform stream die off much more slowly. The flow of water is shown by aluminum dust. Photograph by Sadatoshi Taneda

- In the (very) low Reynolds number limit, the flow is **steady** and **reversible** and very similar to the inviscid potential flow solution

The flow depends on Re

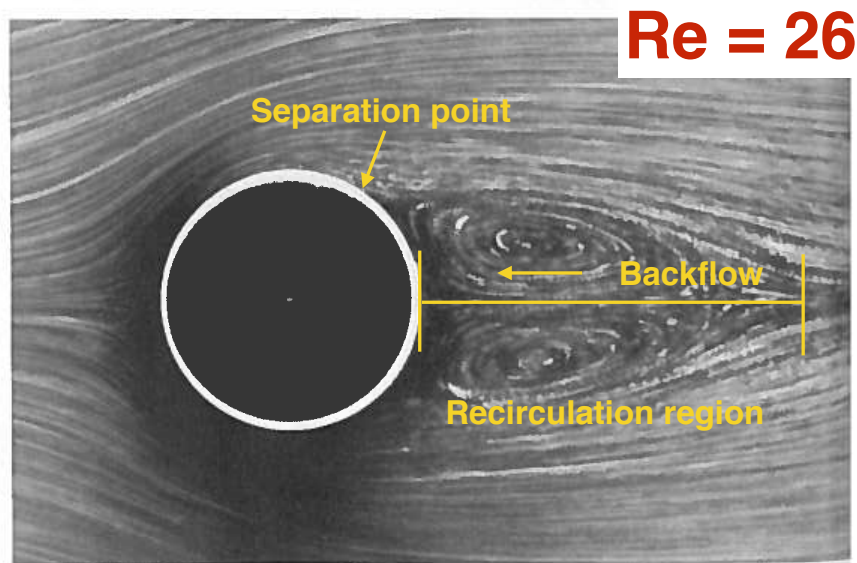


- As the Reynolds number increases, the flow remains **steady** and but is not **reversible** anymore.
- The streamlines are not fore-and-aft symmetric anymore.
- Separation in the wake occurs at about $Re = 5$

24. Circular cylinder at $R=1.54$. At this Reynolds number the streamline pattern has clearly lost the fore-and-aft symmetry of figure 6. However, the flow has not yet separated at the rear. That begins at about $R=5$,

though the value is not known accurately. Streamlines are made visible by aluminum powder in water. Photograph by Sadatoshi Taneda

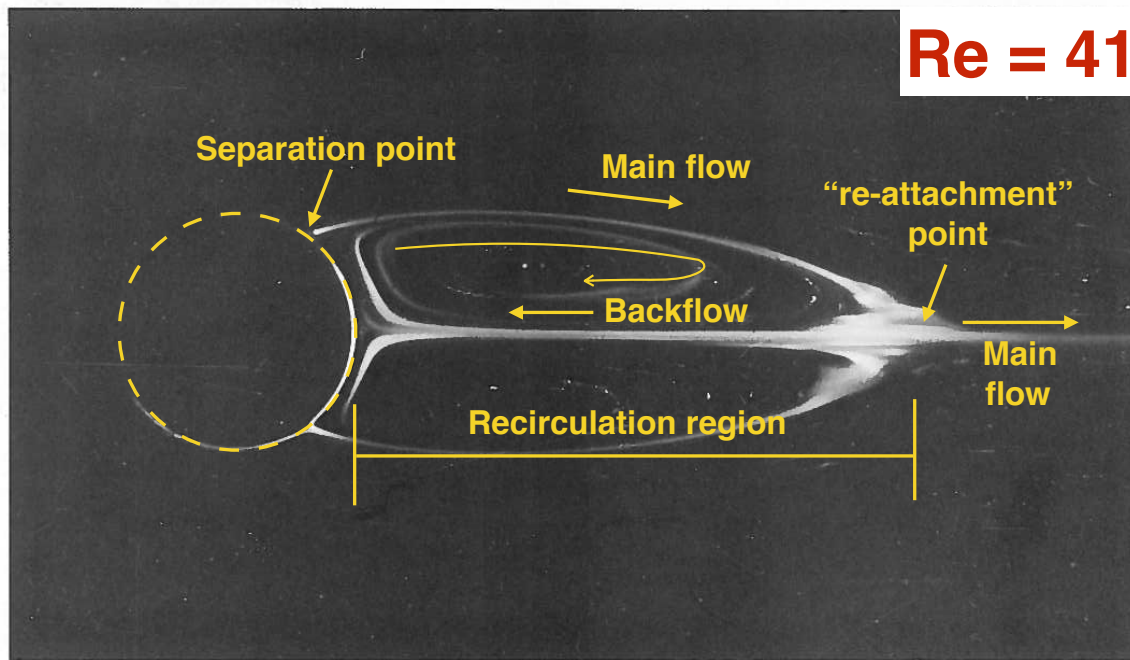
The flow depends on Re



42. Circular cylinder at $R=26$. The downstream distance to the cores of the eddies also increases linearly with Reynolds number. However, the lateral distance between the cores appears to grow more nearly as the square root. Photograph by Sadatoshi Taneda

- At $Re \sim 20$, the flow is **steady** and a **recirculation region** appears in the wake.
- In the recirculation region, the **flow is reversed** and we call the flow “separated”.
- Two large, symmetric “eddy” (flow structures) appear in the wake

The flow depends on Re

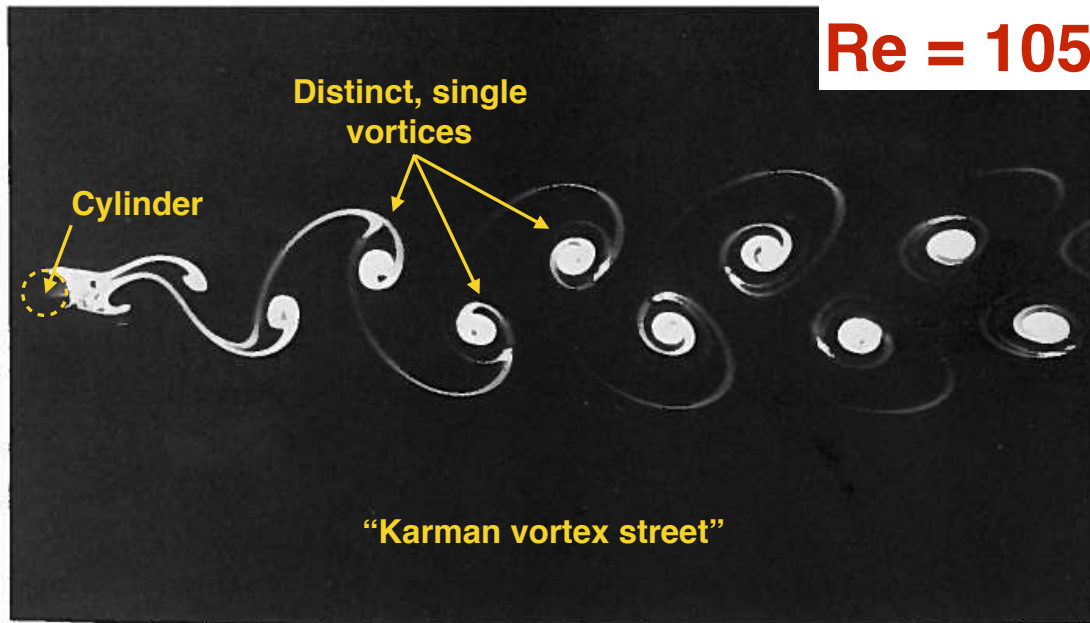


46. Circular cylinder at $R=41.0$. This is the approximate upper limit for steady flow. Far downstream the wake has already begun to oscillate sinusoidally. Tiny irregular

gathers are appearing on the boundary of the recirculating region, but dying out as they reach its downstream end. Taneda 1955

- As the Reynolds increases, **the recirculation length increases.**
- Here the recirculation is more than 2 diameters.
- $Re \sim 40$ is the **upper bound for a steady recirculation region.** Far away, the wake is already oscillating.
- Above this Reynolds, the flow is unsteady and “vortex shedding” occurs.

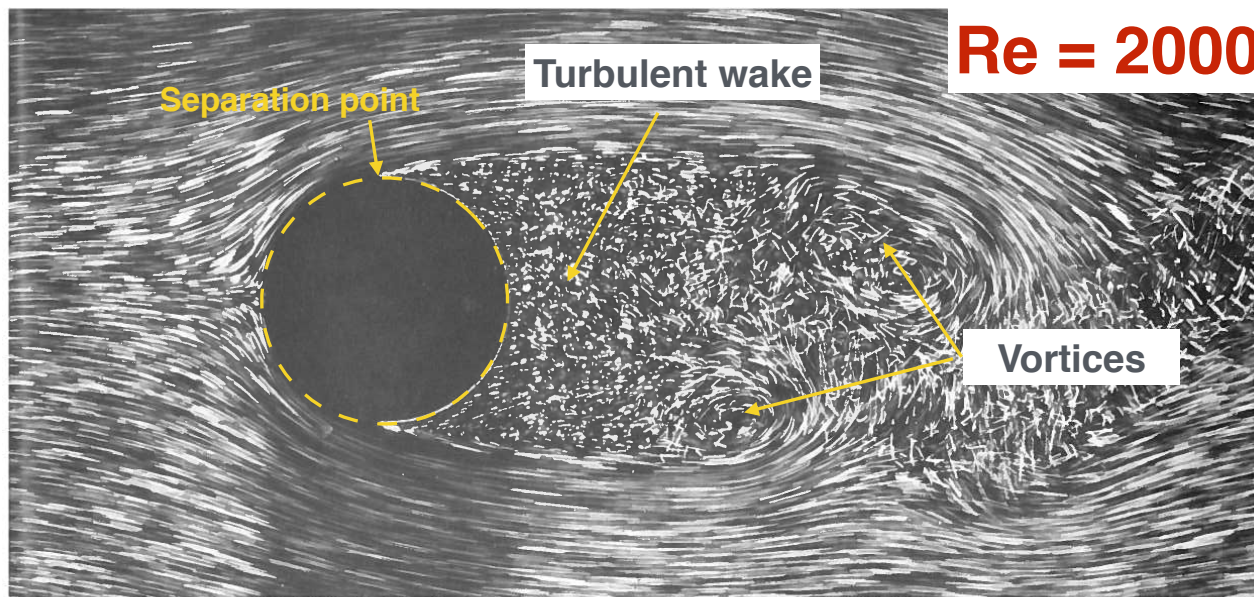
The flow depends on Re



96. Kármán vortex street behind a circular cylinder at $Re=105$. The initially spreading wake shown opposite develops into the two parallel rows of staggered vortices that von Kármán's inviscid theory shows to be stable when the ratio of width to streamwise spacing is 0.28. Streaklines are shown by electrolytic precipitation in water. Photograph by Sadatoshi Taneda

- At Reynolds 100, the “**von Karman vortex street**” is very clear.
- Vortices (regions of large vorticity, surrounded by irrotational flow) are visible and persist for a long distance downstream
- “Shedding” of vortices from the cylinder occurs at a regular frequency.

The flow depends on Re



47. Circular cylinder at $R=2000$. At this Reynolds number one may properly speak of a boundary layer. It is laminar over the front, separates, and breaks up into a turbulent wake. The separation points, moving forward as

the Reynolds number is increased, have now attained their upstream limit, ahead of maximum thickness. Visualization is by air bubbles in water. ONERA photograph, Werlé & Gallon 1972

- At Re 2000, the wake is turbulent.
- Separation occurs on the cylinder's surface earlier than in the laminar case.
- The turbulent wake is much shorter than the last "steady" laminar wake.
- Shedding is still apparent on a background of chaotic, turbulent flow.

Things to note

- We may want to study the flow around an *unconfined* circular cylinder, but the computational domain has finite size (by the way... also a wind- or water-tunnel has finite size...)
 - Boundary conditions *in the far field* for these flows are important
 - Size of the domain is important also. The larger, the more the velocity field will conform to an *unconfined cylinder*, but “how large is large enough?”
- Need to account for steadiness vs. unsteadiness of the flow when simulating the flow.
- Space- and time- resolution requirements for (a) boundary layer on cylinder; (b) separation point/region; (c) time accuracy if interested in vortex shedding